

Question 1:

(10 marks)

Perform Signed Multiplication using Booth's algorithm? Where Multiplicand is 0DFH and Multiplier is

 (89)₁₀?

$$(89)_{10} \xrightarrow{(0DF)_{16}} (11011111)_2$$

$$(89)_{10} \xrightarrow{(59)_{16}} (01011001)_2$$

count

(8)

(7)

(6)

(5)

(4)

(3)

(2)

(1)

(0)

A(Accumulator)	Q (Multiplier)	Q ₋₁	Steps
0 0000 0000	0 0101 1001	0	initialization
1 0010 0001	0 0101 1001	0	A - M
1 1001 0000	1 0010 1100	1	SAR
0 0110 1111	1 0010 1100	1	A + M
0 0011 0111	1 1001 0110	0	SAR
0 0001 1011	1 1100 1011	0	SAR
1 0011 1100	1 1100 1011	0	A - M
1 1001 1110	0 1110 0101	1	SAR
1 1100 1111	0 0111 0010	1	SAR
0 1010 1110	0 0111 0010	1	A + M
0 0101 0111	0 0011 1001	0	SAR
1 0111 1000	0 0011 1001	0	A - M
1 1011 1100	0 0001 1100	1	SAR
0 1001 1011	0 0001 1100	1	A + M
0 0100 1101	1 0000 1110	0	SAR
0 0010 0110	1 1000 0111	0	SAR

Question 2:**(10 marks)**

Perform Unsigned Multiplication using Booth's algorithm? Where Multiplicand is and 071H Multiplier is (218)₁₀?

(For solution)

count	C	Accumulator	Q	Steps
(8)	0	0000 = 0000	1101 1010	initialization
(7)	0	0000 0000	0110 1101	shift
(6)	0	0111 0001	0110 1101	A + M
	0	0011 1000	1011 0110	shift
(5)	0	0001 1100	0101 1011	shift
(4)	0	1000 1101	0101 1011	A + M
	0	0100 0110	1010 1101	shift
(3)	0	1011 0111	1010 1101	A + M
	0	0101 1011	1101 0110	shift
(2)	0	0010 1101	1110 1011	shift
	0	1001 1110	1110 1011	A + M
(1)	0	0100 1111	0111 0101	shift
	0	1100 0000	0111 0101	A + M
(0)	0	0110 0000	0011 1010	shift

Question 3**(5 marks)**

Run following code and update registers accordingly in HEXA?

```
clc
mov ax,0D7BCH
SHL al,2
SAR ax,3
SAL ax,1
SHR al,1
ROL ax,4
RCR al,2
RCL ax,1
ROR ax,8
```

AX	D7BCh
AL	F0h
AX	FAFEh
AX	F5FCh
AL	7Eh
AX	57EFh
AL	FBh
AX	AFF7h
AX	F7AFh